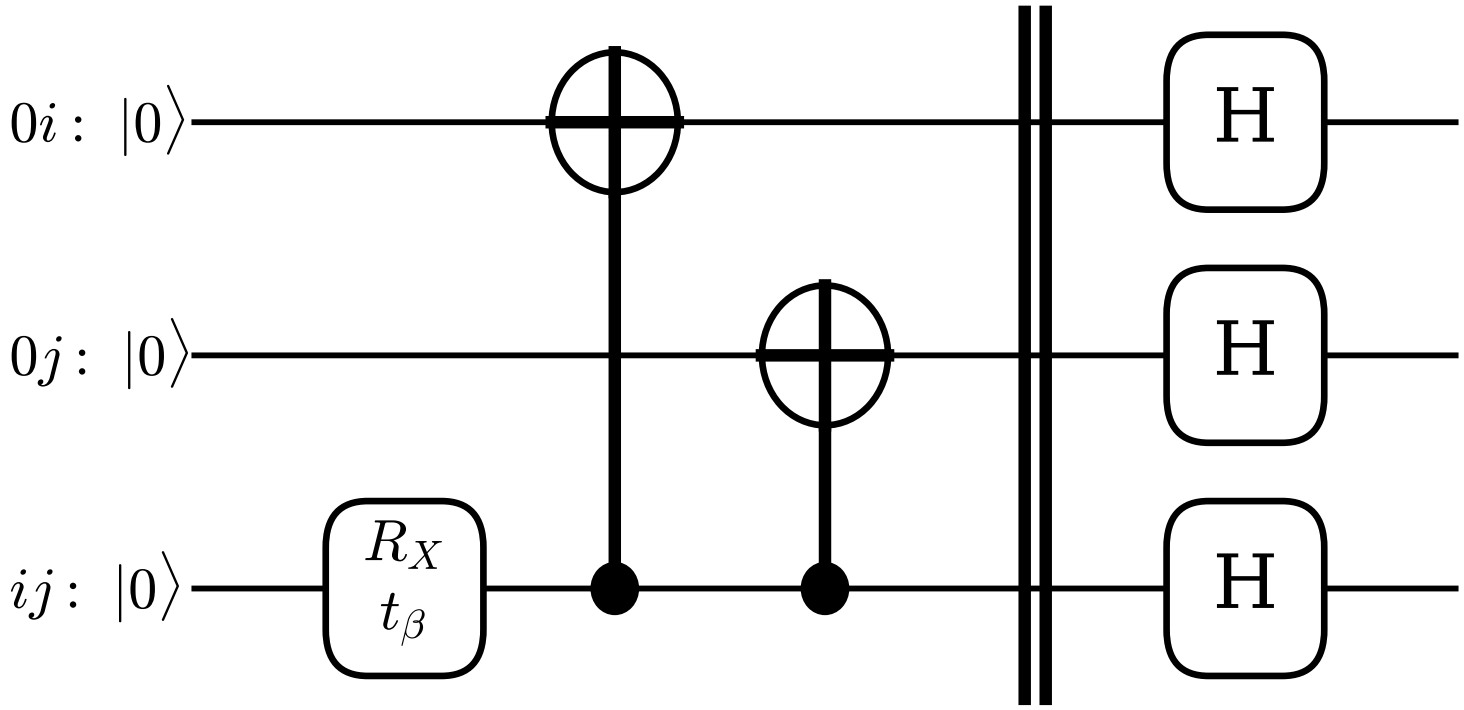
Reference-state preparation

Final Hadamards